

## INVENTORY CENTRAL: USER GUIDELINES

### Inventory Management System Based on Java Swing & PostgreSQL

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#### 1. System Requirements & Installation

##### Software Requirements:

- **Java Development Kit (JDK):** Version 8 or newer.
- **NetBeans IDE:** Version 8.2, 11, or newer.
- **Database Server:** PostgreSQL database server.
- **Database Driver:** PostgreSQL JDBC Driver ([postgresql-42.x.x.jar](#)).

##### Database Installation Steps:

1. Open your **pgAdmin** application or the **PostgreSQL CLI**.
2. Create a new database named [inventory\\_db](#).
3. Execute the DDL script to generate the required table structures: [login](#), [addproduct](#), [cart](#), and [billmain](#).
4. Ensure the default admin credentials are raw inserted into the [login](#) table with **Username:** admin and **Password:** admin.

**How to Open and Run the Application in NetBeans:** If you are running the application directly from the source code rather than using a compiled [.jar](#) file, follow these instructions:

##### *Step 1: Open the Project in NetBeans*

1. Launch your **NetBeans IDE**.
2. Navigate to **File > Open Project....**
3. Browse and select your **Inventory Central** source code folder, then click the **Open Project** button.

*Step 2: Add the PostgreSQL JDBC Driver (Crucial Step!)* To enable the application to establish a secure handshake with your [inventory\\_db](#) database, you must manually link the driver to your NetBeans project environment:

1. Go to the **Projects** panel located on the left-hand side of NetBeans, right-click on the **Libraries** folder under your project hierarchy.
2. Select **Add JAR/Folder...**
3. Browse your local files to find the database driver file you previously downloaded.
4. Click **Open**. Verify that the driver `.jar` file now shows up cleanly inside your project's NetBeans Library list.

### *Step 3: Executing (Running) the Application*

1. Right-click on the root project name (**Inventory Central**) in the *Projects* panel and select **Run**, or simply hit the green **Play (Triangle Icon)** button on the top toolbar.
2. Alternatively, use the built-in keyboard shortcut by pressing **F6**.
3. NetBeans will initiate an automated compilation routine, and the desktop GUI window for the **Loginpage** will promptly launch onto your screen.

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## 2. Getting Started (Login & Registration)

**The Login Screen:** When the application launches, the **Loginpage** interface will greet you.

- **Logging in with Default Account:** Type in `admin` as the username and `admin` as the password, then press the **LOGIN** button.
- **Registering a Brand New User:** If you do not have an active account yet, click the registration prompt to open the **Registeruser** window. Fill out the user details fields provided and hit save.

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## 3. Core Features Guide

Upon a successful login validation, you will instantly be redirected to the **Mainpage**. All application controls are centralized via a modern navigation drawer (sidebar) placed on the far left side.

- **A. Add Product:** Click the **ADD PRODUCT** tab located on the sidebar menu. Fill out the form fields comprehensively (*Item ID*, *Item Name*, *Quantity*, *Price*, *MFD*, *EXP*, and *Batch No*). Click the **ADD PRODUCT** button to commit this new record to the master database, or click **CLEAR** to quickly wipe the form input values clean.
- **B. Inventory (Check Stock Levels):** Click the **INVENTORY** tab on the sidebar menu. Click the **DISPLAY INVENTORY** button to actively query and pull all warehouse item records out of the database. The loaded data is securely generated into a rigid *read-only* table format.
- **C. Update Stock (Manual Stock Adjustments):** Click the **UPDATE STOCK** tab. Key in the *Item ID* of the intended item. The system will instantly fetch and map the corresponding *Item Name* and *Price* data dynamically onto the screen (locked as read-only). Type the additional stock count into the designated input field and click **UPDATE STOCK** to instantly increase the inventory level.

- **D. Track Sales:** Click the **TRACK SALES** tab. Utilize the **DateChooser** drop-down component to pick the precise operational calendar date you want to audit. The interface table will load a granular chronological history of all invoice bills issued on that day.
  - **E. Update Login Details:** Click the **UPDATE LOGIN DETAILS** tab. Provide your current old username and old password, followed by inputting your chosen new username and new password. Click the save button to enforce the changes instantly across the database.
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#### 4. Shopping Cart & Checkout Process (Cart Flow)

1. Click on the **SEARCH PRODUCT** tab from the main sidebar navigation.
  2. Provide the targeted **Item ID** into the search field bar to filter down your inventory.
  3. Left-click directly on the desired row item displayed inside the search table. This action triggers an automatic property mapping, filling the input forms instantly.
  4. Input the client's preferred order volume into the *Quantity* field and click **ADD TO CART**.
  5. Click the **VIEW CART** button to review your pending sales items layout.
  6. Double-check item quantities inside the cart view window and click **CHECKOUT** to finalize the financial transaction. The internal application architecture will automatically subtract those quantities from your warehouse stock and print a randomized 4-digit **Bill Number**.
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#### 5. Sales Returns Management (Return Product)

In the event that an item is defected, incorrect, or requires a formal customer swap out, leverage the following procedural routine:

1. Click on the **RETURN PRODUCT** tab from your main sidebar navigation.
2. Input the exact **Bill Number** (printed on the customer's transaction receipt invoice) and click search.
3. The interface table will query the transaction log and pull up a list of all historical products purchased under that exact bill code.
4. Left-click to select the item row line that needs to be brought back, then fill out the precise numeric count inside the **Return Quantity** column box.
5. Click the return process button. The system automatically re-calculates the **returned\_quantity** logs on that specific transaction profile and safely shifts that returned stock directly back into the warehouse inventory table (**addproduct**).